

TCP Question Bank

Q1. Sequence and Acknowledgement Number

A TCP sender transmits a segment with:

- Sequence Number = 5000
- Data = 1200 bytes

The segment reaches the receiver correctly.

- a) What is the sequence number of the first byte in the segment?
- b) What acknowledgement number should the receiver send?
- c) If the receiver sends a pure ACK, does the ACK itself consume a sequence number? Explain.
- d) If the next segment carries another 800 bytes, what sequence number should it use?

Q2. Missing Segment and Duplicate ACK

A TCP sender transmits three segments:

Segment	Sequence Number	Data Size
S1	1000	1000 bytes
S2	2000	1000 bytes
S3	3000	1000 bytes

S1 reaches the receiver, S2 is lost, but S3 reaches the receiver.

- a) What is the next byte the receiver expects after receiving S1?
- b) What acknowledgement number will the receiver send after receiving S3?
- c) Why does the receiver not acknowledge S3 as successfully received?
- d) What does the duplicate acknowledgement tell the sender about the missing data?

Q3. TCP Three-Way Handshake

A client begins a TCP connection with:

Client → Server: SYN, Seq = 4000

The server responds with:

Server → Client: SYN + ACK, Seq = 8000

- a) What acknowledgement number should the server put in its SYN-ACK?
- b) What sequence number and acknowledgement number should the client's final ACK contain?
- c) Why do SYN segments consume a sequence number even though they contain no application data?
- d) Explain what the three messages establish before actual data transfer begins.

Q4. TCP Data + FIN

Level: Medium–Hard | 10 Marks

A TCP connection has the following event:

A sender transmits 500 bytes of data with Seq = 7000.

After receiving the data, the sender wants to terminate its sending side and transmits a FIN.

- a) What acknowledgement number should the receiver send after correctly receiving the 500 bytes?
- b) What sequence number should the FIN use?
- c) Does the FIN consume a sequence number? Explain.
- d) If the receiver acknowledges the FIN, what acknowledgement number should it send?

Q5. Sequence Numbers Across Multiple Segments

Level: Hard | 10 Marks

A TCP sender starts transmitting data with:

Initial Sequence Number = 10,000

It sends:

- Segment 1 → 1500 bytes
- Segment 2 → 800 bytes
- Segment 3 → 1200 bytes

Segment 2 is lost, while Segments 1 and 3 reach the receiver.

- a) Determine the sequence number of each of the three segments.
- b) What acknowledgement should the receiver generate after receiving Segment 1?
- c) What acknowledgement should it generate after receiving Segment 3 while Segment 2 is still missing?
- d) Explain why the receiver's acknowledgement does not simply advance to the end of Segment 3.

This tests whether the student understands TCP as a byte-stream protocol rather than merely applying a memorized ACK formula.

Q6. Why Does TCP Need a Connection?

TCP is described as a reliable, connection-oriented protocol.

Suppose a student asks:

“If two computers already know each other's IP addresses and port numbers, why can't TCP immediately start sending data?”

- a) What does connection-oriented communication mean in TCP?
- b) Why does TCP need a connection-establishment phase before data transfer?
- c) What role do SYN and ACK play during connection establishment?
- d) Why would simply sending data immediately make reliable communication more difficult?

Q7. “I Received an ACK, So Everything Is Fine”

A TCP sender transmits several segments and receives acknowledgements.

A student says:

“Once an ACK is received, TCP knows that the entire communication is reliable.”

- a) What does a TCP acknowledgement actually indicate?
- b) Why does TCP use sequence numbers along with acknowledgement numbers?
- c) Suppose a segment is lost but a later segment arrives. How can TCP detect that something is missing?
- d) Why is an acknowledgement more useful than simply sending a “received” message?

Q8. Pure ACK vs Piggybacking

TCP can use either a pure acknowledgement or piggybacking.

Consider two hosts communicating in both directions:

A → B: Data
B → A: Data

A student argues:

“Whenever B receives data from A, B should immediately send a separate ACK packet. Otherwise TCP cannot provide reliable communication.”

- a) What is a pure acknowledgement?
- b) What is piggybacking?
- c) Why can piggybacking reduce communication overhead?
- d) Why might TCP still need to send a pure acknowledgement sometimes?

Q9. What Happens When the Receiver Is Slow?

A powerful server is communicating with a slower client using TCP.

The server can generate data much faster than the client can process it.

A student says:

“TCP should simply keep sending as fast as possible because TCP guarantees reliable delivery.”

- a) What problem can occur if the sender transmits data faster than the receiver can handle it?

- b) What is the purpose of TCP's flow-control mechanism?
- c) What information does the receiver provide to help the sender control how much data can be sent?
- d) Explain how the receiver's advertised window can prevent the sender from overwhelming the receiver.

Q10. Why Does TCP Have So Many Flags?

A TCP header contains several control flags, including:

SYN, ACK, FIN, RST, PSH and URG.

A student argues:

“SYN, ACK and FIN are enough. The other flags are unnecessary because TCP already provides reliability.”

- a) What is the purpose of the SYN flag?
- b) What is the purpose of the FIN and RST flags, and how are they different?
- c) Why might TCP need separate control flags instead of representing everything using only sequence and acknowledgement numbers?
- d) Why does the presence of control information make the TCP header more than just a place to store port numbers and sequence numbers?

Network Layer Question Bank

Q1. TTL and Packet Lifetime

A packet is sent from Host A toward Host B through the following path:

Host A → R1 → R2 → R3 → R4 → Host B

The packet starts with TTL = 5. Each router decreases the TTL by 1 before forwarding the packet.

- a) Determine the TTL value after the packet passes through each router.
- b) Will the packet be discarded because of TTL expiry before reaching Host B? Explain.
- c) What would happen if the initial TTL were 3?
- d) What type of ICMP message could the sender receive if the packet is discarded because its TTL expires?

Q2. Finding the Point of TTL Expiry

A packet travels through the following sequence:

Host A → R1 → R2 → R3 → R4 → R5 → Host B

The packet is transmitted with different initial TTL values in three attempts:

- Attempt 1: TTL = 2
- Attempt 2: TTL = 4
- Attempt 3: TTL = 6

Assume every router decreases TTL by 1.

- a) For each attempt, determine at which router the packet will be discarded, if it is discarded.
- b) For which initial TTL value can the packet reach Host B?
- c) What ICMP feedback would the sender receive when TTL expires?
- d) What does repeated TTL expiry at different points suggest about the packet's path?

Q3. ICMP Echo Request and Reply

Host A sends an ICMP Echo Request to Host B. The request travels through three routers before reaching B. Host B sends an Echo Reply back through the same three routers.

Assume the initial TTL of each newly generated IP packet is 8.

- a) What will be the TTL of the Echo Request when it reaches Host B?
- b) What will be the TTL of the Echo Reply when it reaches Host A?
- c) If the Echo Request instead had an initial TTL of 3, what would happen?
- d) Explain what information Host A can and cannot conclude from successfully receiving the Echo Reply.

Q4. Broadcast Address Reasoning

A network administrator wants to send a message to all hosts on a particular network.

The administrator considers the following two approaches:

- Send the packet separately to every host.
- Send one broadcast packet.

Suppose a network contains 50 active hosts.

- a) How many individual transmissions would be required using unicast if the sender sends the message separately to every host?
- b) Conceptually, how does this differ from using a broadcast?
- c) If the administrator wants the message to reach hosts on a particular network rather than only the local network, which type of broadcast should be considered?
- d) Why can broadcasting be more efficient in this situation, and what potential disadvantage could excessive broadcast traffic introduce?

Q5. Diagnosing Different ICMP Outcomes

Host A sends packets toward Host B through a network containing several routers.

During three different attempts, Host A observes:

- Attempt 1: TTL Exceeded
- Attempt 2: Destination Unreachable
- Attempt 3: Echo Reply

- What does each observation indicate about the packet or destination?
- Which observation indicates that the packet's TTL expired?
- Which observation indicates that the destination responded to an ICMP request?
- Can Host A conclude that the entire communication path and all network services are working correctly based only on Attempt 3? Explain.

Q6. Why Does a Packet Need TTL?

A student asks:

“If the destination address of a packet is already known, why should the packet have a TTL? The routers should simply keep forwarding it until it reaches the destination.”

- What problem is TTL designed to prevent?
- Explain what could happen if a packet became trapped in a routing loop and there was no TTL mechanism.
- Why is discarding a packet after TTL expiry preferable to allowing it to circulate indefinitely?
- How does ICMP help the sender understand that TTL caused the packet to be discarded?

Q7. “Destination Unreachable Means the Computer Is Off”

A student sends a packet and receives an ICMP Destination Unreachable message.

The student immediately concludes:

“The destination computer must be switched off.”

- Is this conclusion necessarily correct?
- What does a Destination Unreachable message actually tell the sender?

- c) Give different network-related situations that could result in a destination being unreachable.
- d) Why is it important to distinguish between “the destination is unreachable” and “the destination computer is switched off”?

Q8. Ping Works, So Everything Works?

A student sends an ICMP Echo Request to a remote host and receives an Echo Reply.

The student concludes:

“The network is completely working because ping succeeded.”

- a) What does a successful ICMP Echo Request/Echo Reply demonstrate?
- b) What can the student reasonably conclude about the destination's Network Layer?
- c) What can the student not conclude from a successful ping?
- d) Why is ICMP request/reply useful as a network diagnostic mechanism?

Q9. Unicast or Broadcast?

A university wants to send three different types of information:

1. A message intended for exactly one host.
2. An announcement intended for all hosts on the local network.
3. An announcement intended for all hosts belonging to a particular network.

A student suggests using the same type of addressing for all three cases.

- a) Which type of communication is appropriate for each situation?
- b) Explain the difference between unicast and broadcast.
- c) What is the difference between limited broadcast and directed broadcast?
- d) Why would sending separate unicast packets to every host not always be the best approach?

Q10. Interpreting Network Layer Feedback

Consider the following statement:

“The Internet Protocol attempts to deliver packets, but if something goes wrong, the sender should be informed about the problem.”

A student argues:

“There is no need for a separate error-reporting mechanism. The sender can simply wait and assume that the packet was lost.”

- a) Why is explicit Network Layer feedback useful?
- b) What role does ICMP play in providing such feedback?
- c) Compare the information conveyed by TTL Exceeded, Destination Unreachable, and Parameter Problem.
- d) Why can feedback about a failed packet be more useful for troubleshooting than simply knowing that no response was received?

IP addressing & Subnetting

1. Indicate whether each of the following subnet masks are valid or invalid? For each invalid one, briefly indicate why it is invalid.
 - i) 255.255.255.112
 - ii) 255.255.255.252
 - iii) 255.255.196.0
 - iv) 255.255.32.0
 - v) 255.255.224.0
 - vi) 255.255.254.96
2. Assuming default subnet masks in each case, indicate whether each of the following host address is valid or invalid? For each invalid one, briefly indicate why it is invalid.
 - i) 222.191.63.0
 - ii) 225.160.192.5
 - iii) 127.194.27.131
 - iv) 131.231.245.127
 - v) 191.1.0.0
3. Indicate whether each of the following IP addresses is a valid host address. For each invalid one, briefly indicate why it is invalid.
 - i) 197.15.136.64/26
 - ii) 106.178.35.255/20
 - iii) 165.192.0.63/26
4. For each of the following hosts, determine:
 - the subnet address
 - directed broadcast address for the subnet
 - number of possible hosts on that subnet
 - maximum number of subnets, if the same subnet mask is used for all the subnets in the network (i.e., no variable length subnet masks are used)
 - i) 100.100.100.12/28
 - ii) 125.103.13.127/23
 - iii) 15.17.183.6/21
 - iv) 129.200.50.60/19
 - v) 188.15.110.8/24
 - vi) 87.25.108.12/28
 - vii) 137.156.31.68/19
 - viii) 26.98.102.11/21
 - ix) 104.198.97.28/23
5. For the following subnet masks, determine the corresponding network prefixes.
 - i) 255.255.128.0 /17
 - ii) 255.255.255.240 /28
 - iii) 255.252.0.0 /14
6. For the following network prefixes, determine the corresponding subnet masks.
 - i) /19
 - ii) /23

Data Link Layer :

1. A CRC generator polynomial is x^4+x+1 (10011). For dataword 1101011011, compute the transmitted codeword. Explain why CRC can detect all single-bit and double-bit errors, but not all burst errors longer than the degree of the generator polynomial.
2. Compare the error-detection strength of Internet checksum vs CRC-32 for a burst error of 40 bits. Which is stronger and why? What is the significance of using 1's complement addition instead of simple binary addition.
3. In Hamming code, how many redundant bits are needed for a 7-bit dataword to enable single-bit error correction?
4. In CSMA/CD, derive why the minimum frame size for Ethernet (with 512-bit slot time) is tied to the maximum collision domain diameter.
5. Compare token passing (token ring) with CSMA/CD in terms of performance under high channel load.
6. A MAC address is 48 bits: explain the structural split (OUI vs NIC-specific) and how global uniqueness is enforced.
7. Why can't MAC addresses alone be used for routing across the internet, unlike IP addresses
8. What do the least significant bits of the first octet of a MAC address indicate
9. How does the receiver handle out-of-order frames in Go-Back-N Protocol?
10. Which of the following is not a valid multicast MAC address?
 - a. 01:00:5E:00:00:00
 - b. 01:00:5E:00:00:FF
 - c. 01:00:5E:00:FF:FF
 - d. 01:00:5E:FF:FF:FF
11. A transmission network has a total bandwidth of 100Mbps. If a streaming service requires a Quality of Service (QoS) guarantee of 20% of the total bandwidth, what is the allocated bandwidth for the streaming service ?
 - a. 10 Mbps
 - b. 20 Mbps
 - c. 30 Mbps
 - d. 40 Mbps

12. A Network utilizes a medium access control (MAC) protocol where stations listen before transmitting to avoid collisions. This protocol also incorporates a backoff mechanism for stations that encounter a busy medium. Which of the following protocols best describe this scenario?

- a. Aloha
- b. Token Ring
- c. Carrier Sense Multiple Access (CSMA)
- d. Carrier Sense Multiple Access with Collision Detection(CSMA/CD)

1. The network G has been allotted the classless block 192.168.40.0/24. The network administrator wants to divide this block into 4 equal-sized subnets to accommodate four departments — CSE, ECE, ME, and Admin.

(a) Determine the subnet mask used for this division. For each of the 4 subnets, clearly state the network id, the usable host range, and the directed broadcast id.

(b) Host A has been assigned the IP address 192.168.40.75. Identify which of the 4 subnets Host A belongs to.

(c) A junior network engineer proposes allotting the IP address block 192.168.40.16 – 192.168.40.29 (14 addresses) to a fifth, smaller department. Using the rules governing a valid CIDR block, determine whether this proposal is valid. Justify your answer with respect to each applicable rule.

(d) For each of the following network operations, identify the exact OSI layer responsible for performing it, and justify your answer in one line.

(i) Compressing outgoing data (zipping it) and translating it into a common character set before transmission.

(ii) Splitting a large message into smaller segments at the sender, and demultiplexing an incoming message to the correct destination process among several running processes at the receiver.

(iii) A router examining the destination network id of an incoming packet to decide which outgoing link to forward it on.

(iv) Converting a stream of bits (1s and 0s) into electrical voltage levels for transmission over a copper cable.

(e) Suppose Host A now uses a browser to visit a website hosted on a server in another network, over HTTP. Answer the following short questions.

(i) State the port number used by the DNS protocol, and the port number used by the HTTP protocol.

(ii) Is HTTP a stateful or a stateless protocol? Explain in 1–2 lines what this implies for how the server treats client requests.

(iii) Distinguish between HTTP 1.0 and HTTP 1.1 in terms of the type of connection each uses.